Term Project:

Loads Modeling Using GridLAB-D

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EE 5/430
Analytical Methods for Power Systems

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1 Introduction

A residential distribution transformer often serves several houses whose electrical demand changes throughout the day. Household appliances and electric water heaters do not necessarily operate at the same time. Engineers must consider the combined demand at each instant when selecting a transformer that can serve the houses within acceptable loading and voltage limits.

In this project, you will model three houses supplied by one shared single-phase distribution transformer. Each house will have an electrical load representing household consumption excluding water heating and an explicit electric water-heater model. Supplied household demand and water-draw profiles will drive a full-day simulation. You will use the resulting **coincident demand**[1] to select a transformer from the provided candidate configurations and verify your selection by simulation.

The project connects load modeling and power-flow analysis to an equipment-selection decision. You will examine how ZIP assumptions affect demand, compare measurements at the source and residential services, and justify the smallest candidate transformer that satisfies the project criteria. An IEEE-format paper will present your methods, results, and recommendation. Your conclusion will apply to the simulated day and stated assumptions; it will not establish year-round adequacy.

1.1 Learning Objectives

- Represent household demand and water heating using the appropriate GridLAB-D objects.
- Use time-series data to simulate three houses over a common 24-hour period.
- Explain the effect of ZIP load assumptions on demand and feeder performance.

[1] W. H. Kersting defines "diversified demand" or "coincident demand" as the sum of demands imposed by a group of loads over a particular period.

- Select and verify a transformer using coincident apparent power and voltage limits.
- Communicate an engineering decision using reproducible results.

1.2 Requirements Terminology

- A *shall* specification is mandatory.
- A *should* specification is recommended; follow it or justify an alternative.
- A *may* specification is optional. Optional work receives no separate extra credit and is not necessary for full credit.

1.3 Project Inputs

The course project package will provide the feeder model, three household demand datasets excluding water heating, and three water-draw profiles in gallons per minute. Dataset filenames, units, the assigned day, and candidate details will be identified in the package instructions.

2 Task 1: Model the three houses

Build three residential services on the secondary of the shared transformer. Keep household consumption and simulated water-heating consumption separate so that each component is counted once.

2.1 Required objects and connections

- Each house shall be represented electrically by one triplex_load for household demand excluding water heating and one explicit water heater for water-heating demand.
- Each triplex_load shall be driven by a player using that house's household demand profile. This profile is nominal real-power demand; apply the documented unit conversion and power-factor assumption.
- Each water heater shall receive its house's water-draw profile through a player attached to its water_demand property. The profile is in gpm; it is not an electrical power input.
- Each service shall have a triplex_meter that measures the combined triplex_load and water heater consumption. Both loads shall be connected downstream of that meter.
- The services shall share one single-phase center-tapped transformer using the provided connection model. Phase redistribution is not required.
- If a house object is used to host a water heater, its HVAC and implicit appliance consumption shall be disabled in the starter configuration. No additional household demand shall be added beyond the supplied triplex_load and explicit water heater.
- Each water heater shall initially use an indoor location, a 120 °F setpoint, a 50-gallon tank, a 2 °F thermostat dead band, a 4.5 kW heating element, 60 °F

inlet water, and tank UA of 3.3 Btu/(h·°F). Use the documented initial tank state and verify property units.

- ***Recommended:*** Use a triplex_load as a house demand representation, and a water_heater object to represent water heaters.

2.2 Data and simulation specifications

- Students shall document data units, any missing or invalid values, initial conditions, and software version. Missing input values shall not silently become zero demand.
- All comparisons shall use the same day and initial conditions unless the experiment explicitly changes them. Randomized parameters should be avoided; if used, fix and report the random seed.

2.3 Required results and deliverables

- A working three-house GLM model with its player files.
- A one-line diagram showing the shared transformer, three service meters, and both loads at each service.
- A table of input files, units, and object assignments.

3 Task 2: Measure demand and compare load models

Compare demand at aligned times and explain the differences between input profiles and simulated electrical consumption.

3.1 Measurements and Validation

- The model shall record transformer loading and service voltages, including the line-to-line or line-to-neutral quantity used for each voltage check. Record current at the transformer and residential services.
- For each water heater, students shall record electrical real power and tank temperature along with the input water draw. Explain the relationship between draws, temperature, and electrical demand.
- Students shall plot transformer-input real power and the sum of the three service-meter real powers on the same time axis. The difference represents losses and any other modeled consumption between these measurement boundaries.
- If an additional meter at the feeder source is used, students shall identify the extra lines and transformers included between that meter and the services. Only losses inside the chosen measurement boundary belong in its balance.
- Identify the peak demand of each house and compare it with the transformer peak demand.

3.2 ZIP Comparison Specifications

- The ZIP comparison shall apply to the three triplex_load objects only. Do not replace the explicit water heater models with ZIP loads.

- Students shall run four full-day cases: constant power, constant current, constant impedance, and a mixture of 40% impedance, 30% current, and 30% power. Apply the same mixture to all three household loads within each case.
- The cases shall use the same nominal household real-power profiles and a common 0.95 lagging power factor for their ZIP components. The power factor is a study assumption, not an inferred property of the datasets. Use the appropriate base-power conversion for the selected GridLAB-D properties.
- Water draws, water-heater settings, initial conditions, and network configuration shall remain fixed across ZIP cases. Water-heater electrical output may respond to the changed operating conditions.
- Students shall compare peak coincident apparent power, minimum service voltage, and daily energy across the four cases. Explain how voltage dependence affects the results.

3.3 Required results and deliverables

- Daily plots of household, water-heater, combined service, and transformer demand; related curves may share a figure.
- A source-to-service power comparison and numerical power and energy checks.
- A comparison table for the four ZIP cases and an explanation of the observed differences.

4 Task 3: Select and verify the transformer

Select the smallest transformer candidate that satisfies the project criteria for the three houses. Use the demand study to choose a starting candidate, then verify its performance with the actual transformer configuration.

4.1 Selection specifications

- Students shall use coincident demand. Sum real powers at each timestamp, sum reactive powers at that same timestamp, and calculate the magnitude of the resulting complex power. Do not add separate household peaks that occur at different times.
- The initial estimate shall use the maximum coincident apparent demand observed across the required ZIP cases. State whether the estimate uses transformer-input or secondary demand.
- The selected candidate shall remain at or below 100% of its nameplate apparent-power rating throughout the simulated day in all four required ZIP cases. Use the transformer-input apparent power consistently for this project's loading calculation.
- All monitored service voltages shall remain within 0.95–1.05 p.u. in those cases.

4.2 Verification and recommendation

- Students shall report the selected transformer's peak loading, minimum and maximum service voltage, and estimated total overload duration for each required case. For an acceptable candidate, overload duration is zero.
- Students shall start testing the next smaller candidate, and report the case and time of its first limiting condition. If it also passes, continue downward until the smallest passing candidate is identified.

- A non-convergent run shall be reported separately from a confirmed voltage or loading violation. If no candidate passes, report the evidence and identify the limiting condition rather than changing the criteria.
- Overload duration shall be estimated from the one-minute results with the interval convention stated. Nameplate exceedance shall not be interpreted as a thermal-damage calculation.
- The recommendation may state the chosen rating, its operating margin, and the limitations of assessing one day. No additional growth margin is required by this assignment.
- Students may investigate one water-heater setpoint and a dead band change, keeping the water draws fixed, and assess its effect on demand and tank temperature. Such an extension shall disclose any reduction in hot-water service.

4.3 Required results and deliverables

- The selected transformer configuration and reproducible verification cases.
- A candidate comparison table with rating, ZIP case, peak loading, voltage limits, overload duration, and pass or fail status.
- Plots supporting the selected and next smaller candidate, plus a written sizing recommendation.

5 Submission

Due December 9, 2026. Use the submission time and location posted in Canvas. Submit all three tasks as one project package.

5.1 For Graduate students: IEEE paper specifications

- The paper shall use the IEEE PES conference template and be submitted as a PDF. Target maximum five pages, including figures, tables, and references.
- The abstract shall summarize the objective, method, selected transformer rating, and principal findings.
- The introduction shall explain the engineering purpose and relevant background with citations.
- The methodology shall describe the feeder, load objects, profiles, water-heater parameters, measurement boundaries, ZIP cases, and sizing criteria.
- The results and discussion shall present validation, demand comparisons, and transformer verification. Explain the electrical behavior rather than presenting plots without interpretation.
- The conclusions shall justify the transformer selection and identify the limits of the study.
- Figures and tables shall identify units, measurement locations, cases, and limits. References shall credit data, the starter model, documentation, and other sources.

5.2 For Undergraduate students: Report specifications

- The same specifications listed above; use the report template provided on Canvas.

5.3 Required results and deliverables

- **Graduate Students:** One IEEE-format paper PDF.
- **Undergraduate Students**: One report in PDF.
- A Github repository containing GLM files, input profiles, supporting output data, and analysis or plotting scripts. Python, MATLAB, or a documented spreadsheet may be used for postprocessing.
- A README with the GridLAB-D version, case names, dependencies, and reproduction commands or steps.
- Acknowledge collaboration and external assistance, such as other students, colleagues, or AI (name the AI tool)

5.4 Assessment

- **Model and data setup** — 25%. Correct objects, connections, data handling, and water-heater settings.
- **Validation and load analysis** — 30%. Consistent measurement boundaries, power checks, ZIP comparisons, and interpretation.
- **Transformer selection and verification** — 30%. Coincident-demand reasoning, correct candidate implementation, and evidence supporting the selection.
- Paper/Report and reproducibility — 15%. Clear presentation, citations, complete files, and usable reproduction instructions.

6 Resources

- **Starter exercise:** GridLAB-D Model.
- **IEEE PES template:** IEEE PES Conference Paper Template
  - Download the **LaTeX Template for Conference Papers**